

RESEARCH ARTICLE

Self-Attention-Based Xception Convolutional Neural Network for Keratoconus Stage Classification Using 2-D RGB Corneal Images

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ABSTRACT Keratoconus is an eye disease that can cause significant vision loss if not detected early. The traditional diagnostic procedure uses corneal tomography images, which provide multi-colour-coded maps and quantitative indices describing the rich structural measurements of the cornea. However, obtaining corneal tomography images requires expensive devices, such as the OCULUS Pentacam system. To increase diagnostic accessibility in low-income regions and reduce diagnostic costs, we first proposed using two-dimensional (2D) RGB colour images captured with a smartphone which is a low-cost imaging device. To detect the keratoconus stage using 2D RGB colour images and overcome the limitations of traditional convolutional neural networks, in this study, we developed an attention-based convolutional neural network derived by modifying the Xception architecture for automatic keratoconus stage classification. The proposed network incorporates query-key-value attention modules, an extended middle flow, and depthwise convolutional blocks to better preserve spatial information and enhance channel-wise feature interactions while maintaining a reasonable computational cost. Experiments were conducted on a novel collected labelled dataset comprising normal, mild, and advanced keratoconus cases to validate the proposed attention-enhanced Xception architecture. The proposed model achieved an overall classification accuracy of 92.86% and reported macro-average precision, recall, and F1 scores above 92%, demonstrating consistent performance across the three disease stage classes. Analysis of the confusion matrix demonstrated strong stage-based discrimination, with perfect identification of advanced cases and minimal misclassification between mild and normal stages. These findings demonstrate that the proposed attention-enhanced Xception architecture provides an effective and generalisable solution for early keratoconus screening, particularly in resource-constrained environments where access to advanced ophthalmic imaging systems is limited.

INDEX TERMS Attention mechanism, deep learning, keratoconus detection, medical image classification, Xception.

I. INTRODUCTION

Keratoconus is a progressive visual impairment characterised by thinning and outward protrusion of the cornea, the transparent dome-shaped surface of the eye, into a cone-like shape. The disease usually affects both eyes, although asymmetry between them is common [1]. It typically begins in late adolescence or early adulthood, progresses slowly over several years, and often takes more than a decade to develop. In the early stages, visual deficits can be corrected using

glasses or soft contact lenses. As the condition advances, rigid gas-permeable or scleral lenses are required, and corneal transplantation may be necessary in severe cases. Corneal collagen cross-linking has recently been introduced as an effective treatment to slow or halt disease progression and reduce the need for transplantation [2].

Clinical signs of keratoconus include blurred or distorted vision, increased sensitivity to light and glare, frequent changes in spectacle use, and occasional sudden clouding of vision. Rapid deterioration, particularly in the presence of irregular astigmatism, requires urgent ophthalmological assessment [3]. The exact cause of keratoconus is unknown,

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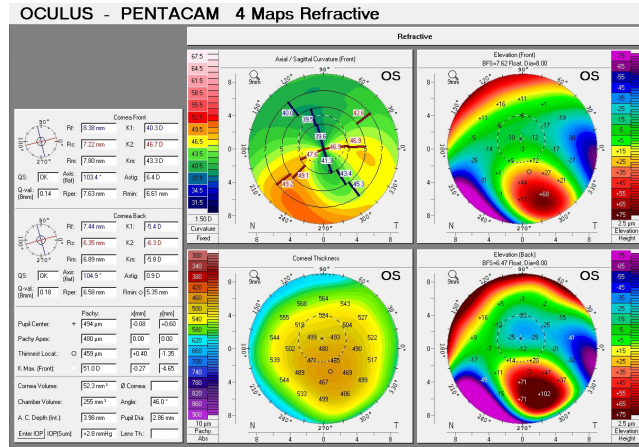


FIGURE 1. 3D corneal tomography map obtained from OCULUS Pentacam system for a single acquisition.

although it is due to a combination of genetic and environmental factors [4]. A family history of keratoconus, vigorous oculorubbing, and various medical conditions, including retinitis pigmentosa, Down syndrome, and skeletal and mucosal disorders, may be related risk factors [5]. Keratoconus can result in complications such as corneal scarring which can progress rapidly, resulting in corneal swelling, diminished vision, sharp pain (hydrops), or slow progressive scarring which may require surgery. Accurate staging of keratoconus is essential for adequate clinical management of this ectatic condition.

A. TOMOGRAPHY-BASED DIAGNOSIS AND PRACTICAL LIMITATIONS

Corneal tomography is one of the most reliable methods for evaluating keratoconus in clinical practice. Devices such as the OCULUS Pentacam use a rotating Scheimpflug camera system to reconstruct high-resolution three-dimensional representations of the cornea and anterior chamber [6]. A single acquisition of a patient generates multiple colour-coded heatmaps and their corresponding quantitative indices which describe the anterior and posterior curvature and corneal thickness of the eye. Figure 1 illustrates the corneal tomography results from the software of the Pentacam device which provides detailed information about the geometry and biomechanics of the cornea. The doctor examined these obtained results with the naked eye to identify corneal abnormalities which is the accepted clinical gold standard. Using such rich information within artificial intelligence models yields near-optimal automated diagnostic performance. However, despite the superior diagnostic accuracy of corneal tomography imaging, the acquisition cost of this device is approximately US\$18,000 and US\$20,000, excluding software modules and maintenance. However, these systems are not financially affordable in resource-limited environments. Consequently, accessing tomography devices is restricted to specialised centres which can cause early keratoconus detection to fail.

B. MOTIVATION FOR LOW-COST RGB IMAGE-BASED SCREENING

In contrast, almost everyone has a smartphone, and modern smartphones are equipped with advanced camera systems which are designed to capture RGB colour images. Smartphones are already used in telemedicine and low-cost health-screening applications. Utilising their accessibility, anterior segment or corneal photographs can be taken without specialised devices and analysed using computer vision and artificial intelligence techniques. The early detection of keratoconus is critical, especially the timely confirmation of corneal tomography and intervention of cross-linking, which can prevent the progression of keratoconus and visual acuity [7]. Therefore, implementing a two-dimensional (2D) RGB colour image-based detection strategy provides a practical and accessible alternative which enables health workers to conduct preliminary in-person screening. However, although conventional convolutional neural networks (CNNs) demonstrate very successful image classification in large-scale datasets, such as the ImageNet challenge, they often struggle to capture subtle and fine-grained morphological variations present in corneal images [8], [9]. This limitation motivated us to develop architectural modifications of the conventional CNN model which is tailored specifically for corneal analysis. Lowering the cost of screening and facilitating early diagnosis helps reduce visual disorders and the economic burden associated with late-stage-related diseases. In this study, we aimed to develop a cost-effective and accessible keratoconus stage detection system using 2D RGB images and a custom CNN model.

C. MOTIVATION FOR MODIFYING XCEPTION NETWORK ARCHITECTURE FOR KERATOCONUS CLASSIFICATION

Over the last decade, image processing and computer vision have been largely driven by deep CNNs. Advanced architectures such as EfficientNet extend earlier designs, including AlexNet, VGG, and Inception [10], [11]. The Xception model is one such architecture, derived from the Inception-V3 concept of depthwise separable convolutions, while retaining VGG-style simplicity to improve computational efficiency and accuracy [12] which is the main reason for selecting the Xception model in this study.

The central idea of Xception is to replace standard convolutions with depthwise separable convolutions by factorising them into spatial (depthwise) and pointwise operations. Although this design is effective for natural images, when applied to medical imaging tasks such as keratoconus classification, aggressive downsampling and simple feature fusion may discard clinically important structural cues. To address these shortcomings, this study introduces targeted architectural enhancements specifically optimised for corneal image analysis.

First, the strided 1×1 convolutions used for downsampling mainly perform linear projection and do not explicitly capture the nonlinear relationships between corneal

biomarkers. Therefore, query-key-value (QKV) self-attention mechanisms were incorporated within the reduction blocks to model long-range spatial and cross-channel dependencies.

Second, conventional pooling or convolution-only reduction operations tend to remove fine spatial details that are essential for detecting the early stages of keratoconus. Depthwise attention blocks were introduced to preserve channel-specific texture information while adaptively emphasising clinically relevant regions.

Third, simple element-wise addition for feature fusion may suppress low-activation but informative responses. Instead, channel-wise concatenation followed by normalisation and 1×1 projection is employed to retain complementary information from both the convolutional and attention branches.

Finally, the original Xception middle flow contains only eight repeated blocks, which may limit the representational capacity. Extending this to 16 refinement blocks increases the effective receptive field and enables richer modelling of progressive corneal deformation patterns.

These modifications were designed to address the following three critical challenges in keratoconus classification:

- (1) preserving micron-scale structural details during feature reduction,
- (2) capturing complex interactions between corneal parameters such as thickness, curvature, and elevation, and
- (3) modelling subtle progression patterns across disease stages.

D. MAJOR ARCHITECTURAL CONTRIBUTIONS

The proposed attention-Xception architecture introduces several structural modifications to the original backbone to better capture the subtle morphological variations that characterise different keratoconus stages.

1) ATTENTION-ENHANCED DOWNSAMPLING BLOCKS

Depthwise convolution is combined with QKV self-attention to preserve channel-specific features while modelling long-range spatial dependencies during feature reduction.

2) IMPROVED FEATURE FUSION STRATEGY

Channel-wise concatenation followed by normalisation and 1×1 projection replaces simple element-wise addition, allowing complementary information from convolutional and attention pathways to be retained.

3) EXTENDED MIDDLE-FLOW DEPTH

The middle flow is deepened from eight to sixteen refinement blocks, increasing the receptive field and enhancing the modelling of both local texture variations and global deformation patterns.

The remainder of this paper is organised as follows. Section II reviews the related literature. Section III describes the RGB corneal dataset. Section IV summarises the original Xception architecture. In Section V, we describe the proposed

attention-based model. Section VI details the experimental protocol. Section VII discusses the results. Finally, Section VIII concludes the paper.

II. RELATED WORK

Recent advances in deep learning have significantly transformed medical image analysis, particularly in ophthalmology. Keratoconus classification remains challenging because of subtle morphological variations that differentiate disease stages [13]. The human cornea exhibits high interindividual variability in curvature, thickness, and elevation, and these nonlinear geometric relationships often complicate automated classification. Consequently, deep learning approaches have been increasingly adopted to directly model intricate spatial dependencies from imaging data. Traditional machine learning methods based on handcrafted features extracted from corneal topography have shown limited generalisability across different imaging devices and acquisition conditions [14]. As a result, research has progressively shifted towards convolutional neural networks (CNNs) and, more recently, attention-based hybrid architectures. In parallel, intelligent optimisation techniques have also been explored to improve feature selection and diagnostic decision-making in medical applications [21].

Anwar and Özbilge [15] for keratoconus detection that integrates YOLOv8-based region-of-interest (ROI) extraction with transfer learning on pretrained CNNs. Unlike earlier studies that relied exclusively on Pentacam tomography, their approach used RGB colour images captured by mobile phones, thereby reducing the cost of keratoconus screening in settings where advanced imaging devices are unavailable. The authors argued that conventional CNNs, when applied directly to full ocular images, introduce redundant background information that reduces classification accuracy. To mitigate this, they applied YOLOv8 to isolate the eye region and classified the cropped images into three stages of the keratoconus disease using five pretrained CNNs (Xception, InceptionV3, ResNet152, InceptionResNetV2, and EfficientNetV2S). Their results demonstrated the benefit of ROI preprocessing: the macro-averaged accuracy over the three classes of the Xception model improved from 91.43% to 93.80%. Their preliminary findings emphasise the importance of targeted preprocessing to improve CNN-based keratoconus detection.

In a comparative study, Ahmed et al. [16] evaluated several pretrained CNN architectures for keratoconus classification. By fine-tuning ImageNet-trained models, such as DenseNet121, EfficientNetB0, InceptionResNetV2, InceptionV3, MobileNetV2, ResNet50, VGG16, and VGG19, on colour-coded corneal topography imaging data, they demonstrated that transfer learning enables strong performance in distinguishing normal, suspect, and keratoconus cases. MobileNetV2 was reported to be the best model with an accuracy of 98% on the test dataset. Their findings confirmed the adaptability of CNN-based feature extractors

to medical imaging applications, particularly when annotated datasets are limited.

Geddada and Pullagura [17] introduced a pyramid-based feature fusion framework using corneal topographic maps obtained from Pentacam imaging. Their dataset included 2,961 images divided into normal, suspect, and keratoconus categories. Multiscale representations were generated using Gaussian, Laplacian, and patch-based pyramids, and histogram of oriented gradient (HOG) descriptors were extracted before feature selection with L1-regularised logistic regression. The tuned LightGBM classifier achieved an accuracy of 85.62% and an AUC of 0.95, with Gaussian pyramid-based features providing the best results. This study reinforces the value of multiscale feature representations in keratoconus detection.

To address the issue of limited datasets, Agharezaei et al. [18] applied Variational Autoencoders (VAEs) for image augmentation. Their dataset of 1,758 corneal topography images was expanded to 4,000 samples using VAE-based generation. Four CNN models, namely, custom-designed CNN, VGG16, EfficientNetB0, and ResNet152V2, were trained on both the original and augmented data. The inclusion of VAE-synthesised images improved the classification performance, with VGG16 achieving 99% accuracy and an AUC of 0.99. This study demonstrated that generative augmentation can effectively address data scarcity in the field of medical imaging.

Recently, Ishrak et al. [19] proposed a hybrid framework that combined CNN feature extractors with Vision Transformers (ViTs) for keratoconus detection using colour-coded corneal topography images. Their approach employed pretrained CNNs, such as DenseNet121, EfficientNetB0, InceptionResNetV2, InceptionV3, MobileNetV2, ResNet50, VGG16, and VGG19, to extract local features, whereas the ViT module captured global dependencies using self-attention. Among the tested combinations, DenseNet121+ViT achieved the highest performance, with 99.28% accuracy in classifying normal, suspect, and keratoconus eyes. Their results showed that hybrid CNN-ViT models can overcome the local receptive field limitations of CNNs and provide scalable and accurate solutions for clinical screening.

Cao et al. [20] comprehensively evaluated eight widely used machine learning algorithms for detecting subclinical keratoconus using Pentacam-derived corneal parameters and demographic features. Their dataset included 49 eyes with subclinical keratoconus and 39 controls, and the models were assessed using 10-fold cross-validation. Random Forest achieved the highest AUC (0.96), Support Vector Machine reached the highest sensitivity (0.92), and k-nearest neighbour provided the best specificity (0.88).

Beyond keratoconus-specific studies, recent computational intelligence methods have also been explored for medical diagnostic optimisation and feature selection. For example, Xiao et al. [21] introduced the Newton downhill optimiser for complex optimisation problems and demon-

strated its application to breast cancer feature selection. This indicates the broader role of intelligent optimisation methods in improving diagnostic-feature selection tasks.

Nearly all of the studies described above rely on corneal tomography images obtained from Pentacam devices, which provide high-quality, standardised, and rich information for training machine learning models. In contrast, Anwar and Özbilge [15] used RGB colour images manually captured from a smartphone, which are subject to greater variability in lighting, resolution, and background noise. Although this makes it difficult and challenging to achieve superior performance for the classifier model, it reduces costs, provides practicality, and enables access to keratoconus screening, especially in environments where specialised imaging devices are not available.

III. MATERIALS - DATASET

A new dataset comprising 420 2D-RGB colour images of the cornea, categorised into three severity levels: normal, mild, and advanced disease, was created to evaluate the proposed keratoconus classification model. These images were collected over three and a half years at the North Eye Center in Iraq under a strict non-disclosure agreement to ensure patient confidentiality. All samples were captured using commercially available mobile devices, such as the iPhone 13 (equipped with a 12.2MP rear camera), enhancing the applicability of the dataset in real-world clinical and resource-limited environments. To the best of our knowledge, this is the only publicly referenced 2D RGB image dataset that provides a complete spectrum of keratoconus severity, making it a valuable contribution to ocular imaging research. The dataset distribution included approximately 35%, 35%, and 30% of mild, advanced, and normal cases, respectively.

Any CNN-based classification model is capable of learning the relationship between the raw input images of the human side profile face and the output of the stage keratoconus disease. However, many attributes in the entire image naturally generate outliers and cause the prediction of the model to deviate from the optimal value. One way to solve this problem is to draw a bounding box in the eye region of the image. Thus, during classification model training, the model focuses only on the extracted ROI, which makes it less sensitive to outlier data points. However, manual ROI extraction is difficult and time-consuming. Therefore, in our previous work [15], we explained that an additional face-side profile dataset was used to train the YOLOv8 model [22] for the automatic extraction of eye regions from raw images. The trained YOLOv8 model was then applied, and each raw image in the keratoconus dataset was replaced with the extracted eye region within the image, as shown in Figure 2.

IV. THE XCEPTION MODEL PARADIGM

This section explains the well-known Xception model, which is the reference architecture used to gradually modify pretrained CNN models to improve their classification

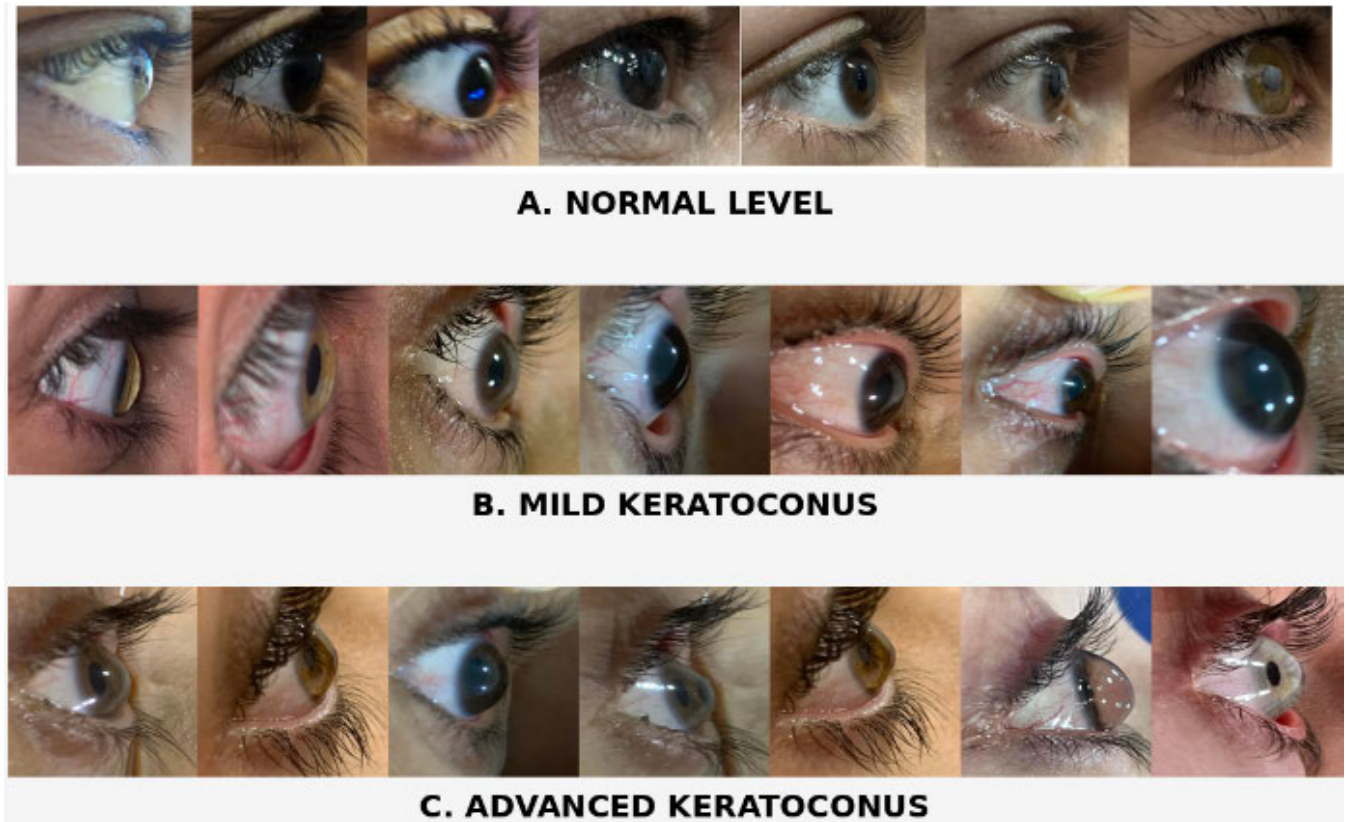


FIGURE 2. Keratoconus samples across advanced, mild, and normal stages.

performance. The introduction of the Xception deep network by Chollet [12] resulted in a significant improvement in the computational efficiency of deep learning. The Xception network comprises three main blocks: entry, middle, and exit flows, as shown in Figure 3. The entry flow begins with standard convolution and max-pooling operations to prepare the feature maps for deeper analysis. This stage is followed by repeated blocks of depthwise separable convolutions, which are the core strength of the model. Depthwise separable convolutions decompose a standard convolution into two sequential operations. First, spatial convolution was applied independently to each channel (*i.e.*, depthwise convolution). Then, a pointwise 1×1 convolution aggregates the channel-wise responses to model the cross-channel interactions, as represented in eqs. (1) and (2).

$$F_{\text{out}} = \text{Conv}_{1 \times 1}(\text{DWConv}_{3 \times 3}(X)), \quad (1)$$

$$\text{DWConv}_{3 \times 3}(X) = \text{Concat}(K_1 * X_1, K_2 * X_2, \dots, K_C * X_C), \quad (2)$$

where $X \in \mathbb{R}^{H \times W \times C}$ denotes the input feature map, K_c is the depthwise kernel associated with the c -th channel, and $\text{Concat}(\cdot)$ concatenates the independently filtered channels along the channel dimension.

The middle flow comprises several identical residual blocks, each containing depthwise separable convolutional layers with ReLU activations, repeated multiple times to

capture increasingly complex patterns while maintaining computational efficiency. In the exit flow, higher-dimensional filters were applied, followed by global average pooling, a fully connected layer, and softmax activation for classification. By relying heavily on depthwise separable convolutions across its structure, Xception reduces the parameter count while preserving the representational power, making it both efficient and effective for large-scale image recognition tasks.

Although the use of depthwise separable convolutions reduces the computational cost, when applied to medical images, particularly in the analysis of corneal topography for subsequent keratoconus classification, the traditional Xception model can become problematic. Max pooling is a standard operation in CNNs without any learnable parameters to reduce the spatial resolution and also condense the feature map to retain highly activated or, in other words, retain dominant feature pixels. However, selecting the maximum activated feature pixel within the corresponding patch causes a loss of relevant information in the feature region, where it stores structural details to distinguish the output class. This is undesirable behaviour, especially in medical image analysis where the detection of subtle morphological variations is essential [23]. Medical imaging modelling requires the preservation of detailed spatial structures while simultaneously selecting relevant features to represent complex anatomical patterns in clinical data. In addition, the Xception network employs linearly activated

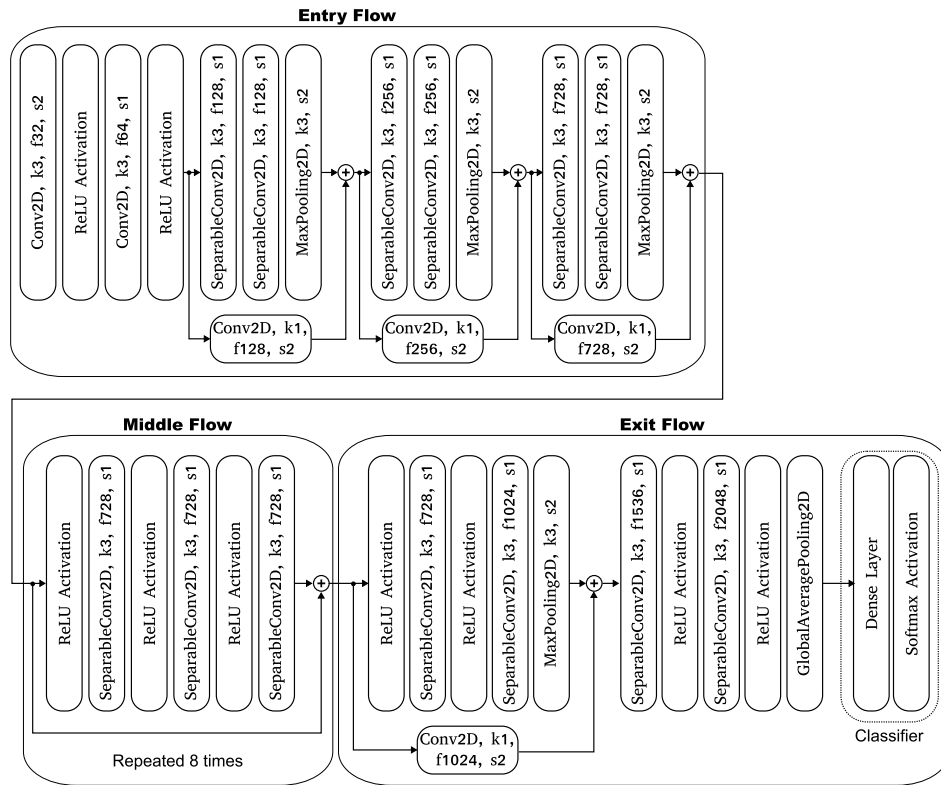


FIGURE 3. Original Xception model architecture.

1×1 convolutions in skip connections within the entry flow. The linear pointwise 1×1 convolution does not take into account the spatial context of the points it processes [24] and fails to capture spatial neighbourhood relationships among biomedical biomarkers. This limitation is particularly critical for diseases characterised by minor morphometric variance, such as keratoconus, where clinically meaningful changes may appear as subtle local deformities.

V. PROPOSED APPROACH

A novel attention-based architecture can provide potential solutions. The fixed pooling and basic projection layers of the Xception architecture can be replaced with adaptive attention mechanisms that automatically readjust dynamically on spatial and channel-wise features based on context. If we shift to context-dependent adaptations, we can achieve greater sensitivity for the early stage detection of keratoconus and cross-platform robustness [25]. To overcome such limitations in the conventional Xception model, this study introduced architectural refinements that address critical gaps in keratoconus classification.

- (1) Pooling-free downsampling that retains fine-scale morphological detail.
- (2) Cross-channel attention mechanisms for modelling complex biomarker interactions.
- (3) An extended middle flow to capture continuous deformation dynamics associated with disease progression better [13], [18], [26], [27].

The following subsections explain the step-by-step modifications of the reference original Xception model, as shown in Figure 3 to yield the proposed attention-based Xception model for keratoconus classification.

A. MODIFIED XCEPTION NETWORK ARCHITECTURE FOR KERATOCONUS CLASSIFICATION

The proposed attention-based convolutional neural network is derived from the Xception model and is organised into four major stages, as illustrated in the block diagrams shown in Figure 4. The first stage corresponds to the entry flow, which is responsible for extracting low- and mid-level representations by jointly exploiting spatial convolution and attention-guided feature modelling techniques. The second stage focuses on learning more discriminative and task-specific features from the representations produced by the entry flow. In the third stage, these features were further refined to encode higher-level patterns that were critical for accurate keratoconus classification. Finally, the last stage aggregates the learned representations using global average pooling, followed by a feedforward classification layer with softmax activation to estimate class probabilities.

An overview of the proposed architecture is shown in Figure 5. While preserving the overall entry-middle-exit flow structure of the original Xception network, several architectural modifications were introduced to enhance feature representation for keratoconus classification. These modifications are summarised as follows:

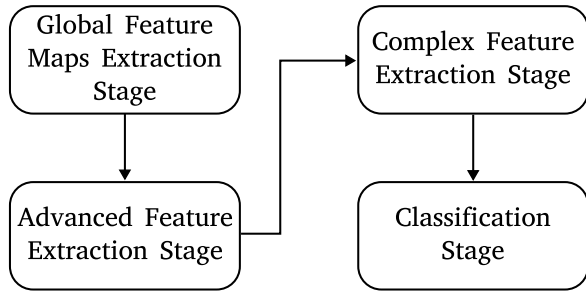


FIGURE 4. Hierarchical feature extraction and classification pipeline.

- (1) Attention-enhanced downsampling blocks;
- (2) Feature concatenation with normalisation;
- (3) Extended middle flow depth.

These design choices are motivated by the nature of medical image analysis, in which subtle morphological variations often distinguish between the disease stages. Unlike the original Xception network, which primarily relies on local convolutional operations, the proposed architecture integrates attention mechanisms to strengthen the cross-channel interactions during feature extraction.

The input image was first processed using two convolutional layers for the initial feature extraction. The first convolution employs 32 filters with a stride of two to reduce the spatial resolution and capture coarse eye-related structures, followed by a second convolution with 64 filters to enrich the low-level representation of the images. The resulting feature map is then passed through three depthwise attention blocks in the entry flow. Each block progressively reduced the spatial resolution while increasing the channel dimensionality, following channel progressions of 128, 256, and 728, as shown in the entry-flow block of Figure 5. The selected channel dimensions were aligned with the corresponding filter sizes used in the entry flow of the original Xception model. This allows us to integrate the proposed attention-based mechanism without dramatically increasing the size of the parameters on the network. The output of the entry flow was forwarded to the middle flow, where the feature maps were further processed using 16 repeated blocks to increase the receptive field and feature depth. Each middle flow block incorporates a skip connection to mitigate gradient degradation and stabilise the training [28]. The refined representations are subsequently passed on to the exit flow, which performs the final stage of hierarchical feature consolidation and produces the class predictions. The architectural components introduced at each stage are described in detail in the following sections.

B. DEPTHWISE ATTENTION BLOCK FOR FEATURE REDUCTION AND FUSION

In the original Xception network, feature map reduction is achieved by processing the input through two parallel branches and combining their outputs using element-wise addition. In this structure, feature reduction is performed using strided 1×1 convolution in the residual branch

and max pooling in the main branch (see the entry flow of the Xception network in Figure 3). Although effective, this strategy relies primarily on local operations and may discard subtle channel-specific information during aggressive downsampling. In the proposed attention-Xception model, this reduction mechanism is redesigned while preserving the two-branch processing paradigm, resulting in a depthwise-attention block that jointly performs spatial reduction and contextual feature enrichment, as shown in Figure 5.

Specifically, the depthwise separable (depthwise and pointwise) convolution used in the original Xception architecture was decomposed such that only the depthwise convolution component was applied to the main processing branch. This design choice is consistent with recent convolutional network architectures, such as ConvNeXt [29], which adopt depthwise convolution as the core operation in the modern CNN backbone. In this configuration, each input channel is independently convolved with its own filter, thereby avoiding channel-wise summation and preserving the channel-specific representations. Spatial downsampling was achieved by applying a stride of two during depthwise convolution, reducing the feature map resolution from $H \times W$ to $H/2 \times W/2$.

The output of the depthwise convolution is subsequently enhanced using a multi-head self-attention mechanism that learns long-range dependencies across spatial locations and channels and assigns adaptive importance weights to the feature responses. Given the depthwise-convolved feature map $X_{dw} \in \mathbb{R}^{H \times W \times C}$, the query Q , key K , and value V tensors are obtained via learnable 1×1 convolutional projections as

$$Q = W_q * X_{dw}, \quad (3)$$

$$K = W_k * X_{dw}, \quad (4)$$

$$V = W_v * X_{dw}, \quad (5)$$

where $W_q, W_k, W_v \in \mathbb{R}^{1 \times 1 \times C \times F}$ are learnable projection weights, and $*$ denotes the convolution. The output dimensionality F was selected to match the channel widths of the corresponding stages in the original Xception architecture, namely, 128, 256, and 728 in the entry flow, and 1024 in the exit flow.

The attention operation follows the scaled dot-product formulation [26] as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (6)$$

where d_k denotes the dimension of the key vector. The attention output is then projected back to the feature space and normalised using layer normalisation (LN) to obtain the attention-enhanced feature map

$$F_{\text{attn}} = \text{LN}(W_p * \text{Attention}(Q, K, V)), \quad (7)$$

where W_p denotes a learnable 1×1 projection. Layer normalisation is preferred at this stage because self-attention

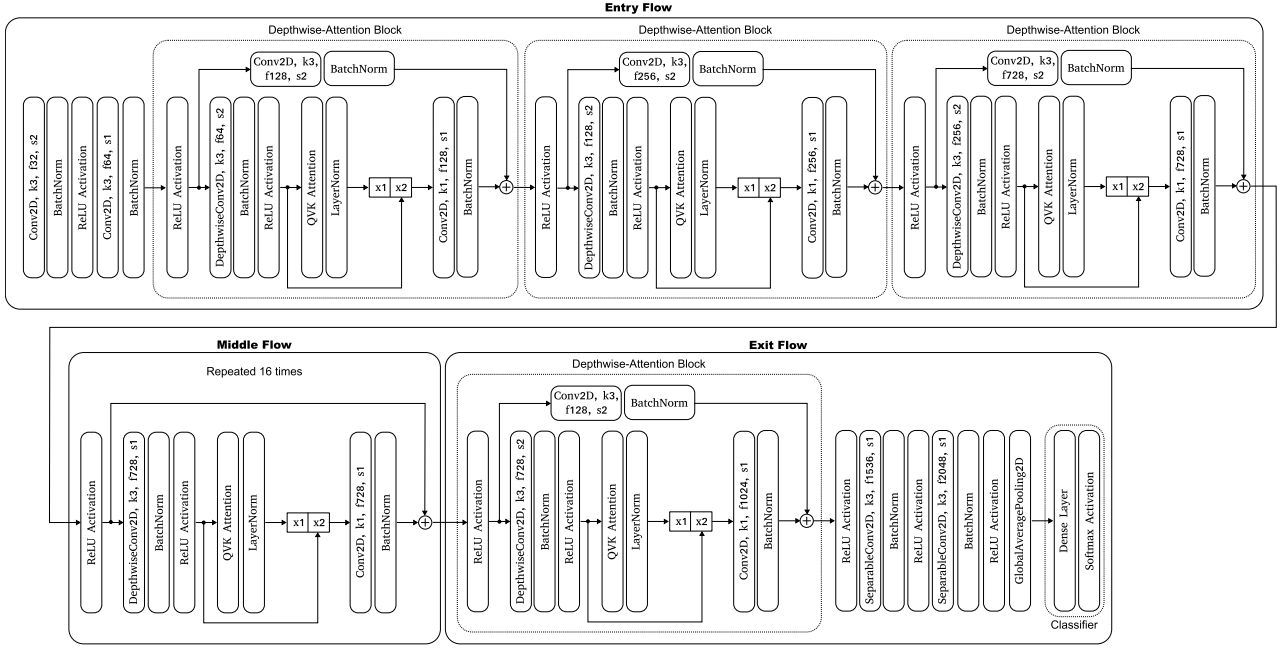


FIGURE 5. Proposed attention-based Xception model architecture.

operates independently of the batch dimension, and LN stabilises feature statistics across channels for each sample.

In parallel with the attention-enhanced branch, the second branch performs feature reduction using a conventional strided convolution to ensure direct transformation. Instead of combining these branches using element-wise addition, as in the original Xception design, the proposed architecture employs channel-wise concatenation. Specifically, depthwise-convolved and attention-enhanced features were concatenated along the channel dimension. This channel-wise concatenation differs fundamentally from the summation-based fusion used in the original Xception network because it preserves complementary information from both branches and explicitly increases channel dimensionality during spatial downsampling. The subsequent 1×1 convolution enables channel mixing and compression of concatenated features to the desired dimensionality while preserving the spatial resolution. Using a two-branch structure with concatenation and residual pathways also allows effective gradient propagation from deeper layers to earlier feature representations which stabilises training and prevents the gradient signals from vanishing through repeated downsampling operations in the depthwise attention blocks of the entry flow. The flow of functional operations of the depthwise attention block is provided in eqs. (8) to (10), where DWConv denotes depthwise convolution, BN denotes Batch Normalisation, and Concat denotes channel-wise concatenation.

$$F_{dw} = \text{BN}(\text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{DWConv}_{3 \times 3}(X))))), \quad (8)$$

$$F_{\text{concat}} = \text{Concat}(F_{dw}, F_{\text{attn}}), \quad (9)$$

$$F_{\text{out}} = \text{BN}(\text{Conv}_{1 \times 1}(F_{\text{concat}})). \quad (10)$$

C. EXTENDED MIDDLE FLOW

The middle flow of the original Xception network was designed to progressively extract more complex feature representations while preserving the spatial resolution of the input feature map from the entry flow. In the proposed attention-Xception architecture, this design principle is retained; however, the middle flow is extended to enhance the representational capacity of the network. Specifically, the number of repeated middle-flow blocks was increased from eight to 16, allowing for deeper feature refinement without modifying the spatial dimensions of the feature maps.

Unlike the depthwise-attention blocks employed in the entry and exit flows, the extended middle-flow blocks do not perform spatial downsampling. The spatial resolution and channel dimensionality of the input feature maps remained constant within each block. Each middle-flow block consists of a main feature refinement path and a direct identity skip connection. The input feature map was first processed through depthwise (group) convolution, followed by BN and ReLU activation, whereas the original input was preserved through an unmodified skip connection.

After depthwise convolution and nonlinearity, the refined features were passed through a multi-head self-attention module. This attention operation captures long-range dependencies across spatial locations and channels and expands the intermediate feature representation. A residual connection was applied around the attention module to stabilise the training and preserve the original feature context. Because the concatenation of the attention-enhanced features with the depthwise-convolved features results in an expanded channel

dimensionality, a 1×1 convolution is employed to project the representation back to the original number of channels. Finally, the projected features are combined with the identity skip connection via an element-wise addition.

The functional operations of a single extended middle-flow block can be expressed as

$$F_{\text{res}} = X, \quad (11)$$

$$F_{\text{dw}} = \text{ReLU}(\text{BN}(\text{DWConv}_{3 \times 3}(X))), \quad (12)$$

$$F_{\text{attn}} = F_{\text{dw}} + \text{Attention}(F_{\text{dw}}), \quad (13)$$

$$F_{\text{concat}} = \text{Concat}(F_{\text{dw}}, F_{\text{attn}}), \quad (14)$$

$$F_{\text{proj}} = \text{Conv}_{1 \times 1}(F_{\text{concat}}), \quad (15)$$

$$F_{\text{out}} = F_{\text{proj}} + F_{\text{res}}. \quad (16)$$

where F_{res} denotes the identity residual feature map and F_{out} represents the output of the middle-flow block.

By extending the depth of the middle flow while maintaining full spatial resolution, the proposed design enables the network to model increasingly complex corneal deformation patterns at all stages of keratoconus. Increasing the number of middle-flow blocks from eight to 16 constitutes one of the major architectural modifications contributing to the increased model complexity of the proposed network. As a result of the architectural changes introduced in the proposed architecture, the total number of trainable parameters of the complete network, including the classification head, increased from 22.1M to 24.9M. This increase in model capacity improved the ability of the network to capture subtle and progressive structural variations. The structure of the extended middle-flow block is shown in Figure 5.

D. EXIT FLOW AND CLASSIFICATION

The exit flow receives the final feature representation produced by the middle flow (see Figure 5). At this stage, the feature map is processed using the same depthwise-attention block employed in the entry flow, following an identical architectural configuration as that of the entry flow. As discussed in the previous subsection, this block jointly performs spatial reduction while increasing the feature depth, resulting in a richer and more discriminative representation of the input image. In the exit flow, this operation increased the number of feature channels to 1024, thereby enabling the network to encode higher-level semantic patterns prior to classification.

After the depthwise-attention block, the resulting feature map was further refined using two depthwise separable convolution blocks. Subsequently, a Global Average Pooling (GAP) layer is applied to collapse the remaining spatial dimensions into a compact one-dimensional feature descriptor. In the proposed architecture, the GAP operation produces a 2048-dimensional feature vector that is forwarded to a fully connected classification layer. The classifier performs a linear transformation on this feature vector, and the resulting logits are normalised using a softmax function to obtain probability

scores across the target classes. The complete computation from the GAP output to the final softmax probabilities is given eqs. (17) to (19).

$$\mathbf{z} = \text{GAP}(X), \quad \mathbf{z} \in \mathbb{R}^{2048} \quad (17)$$

$$\mathbf{o} = W\mathbf{z} + \mathbf{b} \quad (18)$$

$$\hat{y}_i = \frac{\exp(o_i)}{\sum_{j=1}^C \exp(o_j)}, \quad i = 1, \dots, C \quad (19)$$

where X denotes the obtained feature map before the global average pooling layer, W represents the weight matrix of the fully connected layer, $\mathbf{b}$ is the corresponding bias vector, $\hat{y}_i$ denotes the predicted probability for class i , and C indicates the total number of classes (*i.e.*, normal, mild, and advanced keratoconus).

VI. EXPERIMENTAL PROCEDURE

The training of the attention-Xception network architecture was conducted with careful consideration of the challenges inherent in medical image classification, including the limited dataset size and class imbalance. This section describes the image augmentation methods, training protocol, and hyperparameters, and provides evaluation metrics for the model assessments.

A. IMAGE AUGMENTATION

The collected keratoconus dataset which contains 420 eye images, was divided into training (60%), validation (10%), and testing (30%) sets for the purpose of this study. For the primary experimental protocol, a fixed train-validation-test split strategy was adopted for model development, hyperparameter tuning, and final evaluation. All images were resized to $[299 \times 299]$, the same as the input image size of the original Xception model, so that we could directly compare the parameter increase in both attention-Xception and original Xception models. To increase generalisation and reduce overfitting, various data augmentation methods were applied to the training set before model training. These methods include (i) random contrast adjustment, (ii) image rotation, (iii) linear translation, (iv) zooming, and (v) horizontal and vertical flipping. Each augmentation method was applied probabilistically in each image ten times to generate diverse variants. Importantly, the original images were also included in the training set which was unaltered without any augmentation methods applied to provide a clear reference for keratoconus. Therefore, we obtained 14,700 augmented images and 294 original images for a total of 14,994 images in the training set.

B. MODEL TRAINING

To mitigate overfitting and ensure stable convergence, the following hyperparameters and strategies were adopted: The Adam optimiser was selected for its adaptive learning rate capabilities and efficiency in managing sparse gradients. The initial learning rate was set to $\eta = 1 \times 10^{-4}$. A dynamic learning rate reduction was employed to refine weight updates

during training. The learning rate was reduced by a factor of 0.5 if the validation loss failed to improve for five consecutive epochs, and a lower bound was set at $\eta_{\min} = 1 \times 10^{-6}$ beyond which no further reduction was performed. A moderate batch size of 16 was employed. This size represents a balance between the stability of gradient estimates (favoured by larger batches) and the regularising effect of noise in the gradient updates (favoured by smaller batches) [30]. In this study, the QKV attention layers were configured with a single header ($h = 1$). One reason for using a single header is to avoid increasing the computational complexity owing to the power source limitations. The proposed attention-Xception model was trained from random initialisation using the default Glorot uniform initializer for convolutional and dense kernels. ImageNet-pretrained weights could not be directly transferred to the proposed architecture because the introduced depthwise-attention blocks, feature concatenation operations, and extended middle flow changed the layer structure and parameter dimensions. Therefore, the baseline pretrained CNNs benefited from available ImageNet initialisation, whereas the proposed model was trained without external pretraining. In addition to the inherent regularisation provided by batch normalisation and data augmentation [31], a weight decay (L2 regularisation) factor of $\lambda = 1 \times 10^{-4}$ was applied to all kernel weights to prevent over-complexity and improve generalisation. A maximum of 200 epochs were used during training. To prevent overfitting, an early stopping strategy with a patience of 10 epochs was employed by monitoring the validation loss and restoring the model weights from the best performing epoch. In addition, a model checkpointing strategy was applied, in which only the best-performing model was saved during training. The focal loss cross-entropy function ($\mathcal{FL}$) is used during model training, where this loss function has an advantage over the standard cross-entropy loss when the distributions of the data in each class are imbalanced [32]. The focal loss function is defined as follows:

$$\mathcal{FL} = - \sum_{i=1}^C \alpha_i y_i (1 - \hat{y}_i)^\gamma \ln(\hat{y}_i), \quad (20)$$

where C indicates the number of output classes; y_i is the ground-truth label for class i ; $\hat{y}_i$ is the predicted softmax probability for class i ; γ is the focusing parameter; and α_i is the weighting factor, which is set for each class independently using the inverse class frequency to overcome the class imbalance problem.

Table 1 summarises the architectural and training hyper-parameters used for the proposed attention-Xception model, together with the tested search space configurations.

C. PERFORMANCE EVALUATION METRICS

The model performance was rigorously evaluated on the test set using several metrics that provided insights into both the overall classification performance and the per-class behaviour. A confusion matrix was used to highlight the

misclassifications made by the model. This is particularly important in clinical applications, where the cost of false negatives (missing a disease) and false positives (misdiagnosing a healthy patient) can be high. To evaluate the performance of the models, several statistical metrics were computed, including accuracy (ACC), precision (PRE), recall (REC), F1 score, and specificity (SPE), as given in eqs. (21) to (25).

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (21)$$

$$\text{PRE} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (22)$$

$$\text{REC} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (23)$$

$$\text{F1} = \frac{2 \cdot \text{PRE} \cdot \text{REC}}{\text{PRE} + \text{REC}} \quad (24)$$

$$\text{SPE} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (25)$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives respectively.

These metrics were computed separately for each class, whereas the macro-averaged precision, recall, and F1-score were additionally reported to summarise the overall classification performance. In contrast, the reported overall accuracy was computed globally using the accumulated TP, TN, FP, and FN values across all classes, rather than by averaging the individual class-wise accuracies.

VII. EXPERIMENTAL RESULTS

The training and validation results for the model accuracy and loss are summarised in Figure 6. It can be observed from the plots that the model gradually reaches an accuracy of approximately 95% on the training dataset and approximately 90% on the validation set. Similarly, the training loss converged to approximately 0.1, whereas the validation loss stabilised at approximately 0.4 for the multi-class classification task. The accuracy curves exhibit a steady improvement throughout training and eventually converge to a stable plateau, while the corresponding loss curves decrease consistently without large oscillations. It is also observed that the validation accuracy and loss curves begin to stabilise after approximately 50–75 epochs, indicating that the proposed attention-Xception model achieves convergence within a relatively limited number of training epochs. These observations demonstrate a stable optimisation process and indicate that the proposed architecture achieved reliable convergence during training. The gap between the training and validation losses indicates a moderate level of variance; however, this behaviour is expected because the model was trained on raw mobile phone RGB colour images captured under different environmental conditions. Differences in illumination, camera distance, focus, and background contributed to the distribution shifts between the training and validation samples, which also

TABLE 1. Architectural and training hyperparameter configurations used for the proposed attention-Xception model.

Category	Hyperparameter	Value/Tested Range
Architecture	Number of Filters	{32, 64, 128, 256, 728, 1024, 1536, 2048}
	Activation Function	ReLU
	Number of Attention Heads	{1, 2, 4}
	Filter Sizes	3×3 , 1×1
Training	Initial Learning Rate	1×10^{-4}
	Minimum Learning Rate	1×10^{-6}
	Learning Rate Scheduler	ReduceLROnPlateau
	Optimizer	Adam
	Weight Initialisation	Glorot uniform random initialisation
	Batch Size	16
	Input Image Size	$299 \times 299 \times 3$
	Weight Decay (L2 Regularisation)	1×10^{-4}
	Loss Function	Focal Loss
	Checkpoint Strategy	Best validation checkpoint saving
	Early Stopping	Enabled with patience of 10

explained the fluctuations in the validation loss curve. Nevertheless, the relatively small separation between the training and validation curves suggests that the model maintained good generalisation capability without exhibiting significant overfitting.

Figure 7 illustrates the evolution of the precision, recall, and F1-score for both the training and validation sets. In the early stages of training, the validation metrics exhibited noticeable fluctuations and remained below the training curves, indicating that the model initially struggled to generalise. As training progressed, the validation precision, recall, and F1-score gradually stabilised and approached the corresponding training values. This convergence suggests that the network can adapt to the variability inherent in the input images and ultimately achieve a stable and reliable generalisation performance.

A quantitative performance comparison between the proposed attention-based Xception model and five commonly used pretrained CNN architectures, namely EfficientNetV2S, InceptionResNetV2, InceptionV3, ResNet152, and the standard Xception network, is presented in Table 2. It is important to note that these models were initialised with ImageNet weights which were not trained from scratch, which means they had already learned some knowledge about eye features because of the large-scale ImageNet dataset. This resulted in less sensitivity at the beginning of the training, unlike the attention-Xception model which was trained from scratch using randomly initialised connection weights. The baseline models achieved balanced precision and recall values in the range of approximately 83–90%, demonstrating that pretrained CNNs can provide a reasonable starting point for keratoconus staging when applied to mobile phone images.

Among these, InceptionResNetV2 yielded the best baseline performance with an accuracy of 90.48%.

In contrast, incorporating the attention mechanism into the Xception architecture consistently improved all the evaluation metrics. The proposed attention-Xception model achieved the highest overall accuracy of 92.86%, with a matching recall of 92.86%, precision of 92.99%, F1-score of 92.85%, and specificity of 96.43%. These results indicate that the attention mechanism improves the sensitivity to keratoconus cases and the ability to reduce false-positive predictions compared with the baseline CNN architectures.

From a model complexity perspective, the proposed attention-Xception network achieves this performance gain with only a modest increase in the parameter count. As shown in Table 2, the attention-Xception model contains 24.9M parameters, which is comparable to Xception (22.1M) and InceptionV3 (22.9M), and substantially smaller than InceptionResNetV2 (55.2M) and ResNet152 (59.5M). Despite having less than half the number of parameters of these deeper architectures, the proposed model achieved superior classification performance. This indicates that the introduced attention and feature fusion mechanisms improve the representational efficiency, enabling more discriminative feature learning without relying on an excessive model capacity.

Table 3 presents the class-wise performance of the proposed attention-based Xception model for normal, mild, and advanced stages of keratoconus. The advanced class achieved perfect performance across all metrics, including 100% accuracy, recall, precision, F1-score, and specificity, indicating that advanced keratoconus cases were consistently

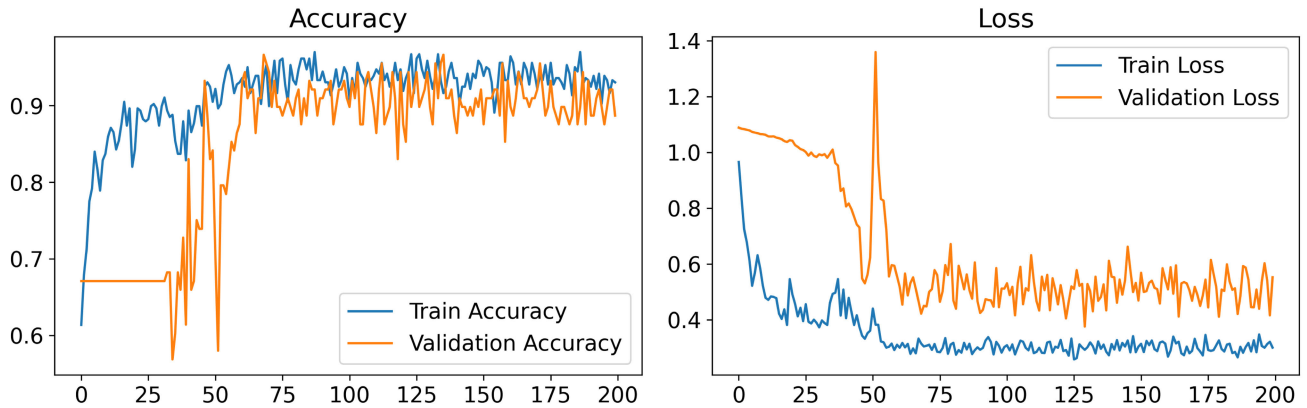


FIGURE 6. Training and validation accuracy and loss curves illustrating the convergence behaviour and optimisation stability of the proposed attention-based Xception model.

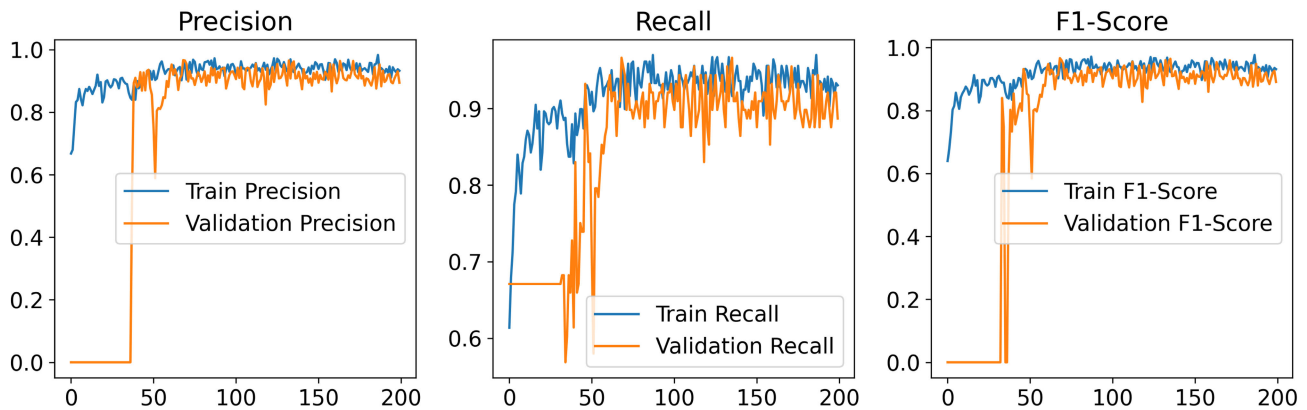


FIGURE 7. Training and validation precision, recall, and F1-score curves demonstrating the generalisation performance of the proposed attention-based Xception model during training.

TABLE 2. Overall performance comparison of the proposed attention-based Xception model with baseline pretrained CNN architectures.

Model	Overall Performance Metrics					#Params
	ACC (%)	REC (%)	PRE (%)	F1 (%)	SPE (%)	
EfficientNetV2S	88.89	88.89	89.18	89.00	94.44	21.1M
InceptionResNetV2	90.48	90.48	90.40	90.43	95.25	55.2M
InceptionV3	84.92	84.92	86.03	85.20	92.46	22.9M
ResNet152	82.54	82.54	83.44	82.86	91.27	59.5M
Xception	88.10	88.10	88.23	88.09	94.05	22.1M
Attention-Xception	92.86	92.86	92.99	92.85	96.43	24.9M

and reliably identified. For the normal and mild classes, the accuracy remained at 92.86%, with F1-scores of 89.66% and 88.89%, respectively. Mild keratoconus presents greater classification difficulty, which is reflected in its lower recall than that of precision. This behaviour is expected because early or mild keratoconus often exhibits subtle morphological

patterns that overlap with those of normal corneas in the RGB images.

The confusion matrix shown in Figure 8 provides additional insight into the observed error patterns. All advanced keratoconus cases were correctly classified, which is consistent with the perfect recall and precision values reported for

TABLE 3. Per-class performance metrics of the proposed attention-based Xception model for normal, mild, and advanced keratoconus stages.

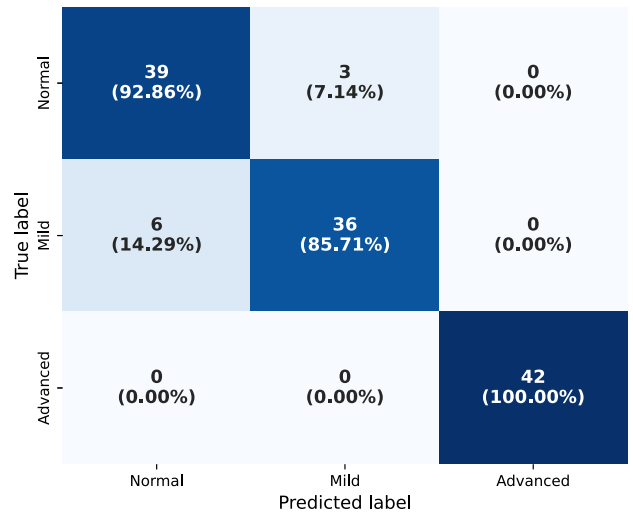
Class	Per-Class Performance Metrics				
	ACC (%)	REC (%)	PRE (%)	F1 (%)	SPE (%)
Normal	92.86	92.86	86.67	89.66	92.86
Mild	92.86	85.71	92.31	88.89	96.43
Advanced	100.0	100.0	100.0	100.0	100.0

the advanced class. The primary source of misclassification occurred between the normal and mild stages of the disease. Some mild cases were predicted to be normal, whereas a small number of normal samples were predicted to be mild. This behaviour is reasonable because early keratoconus may present only subtle visual cues in RGB images, making it difficult to distinguish it from healthy corneas. Importantly, no normal or mild cases were misclassified as advanced, indicating that the model maintained a clear separation between early stage and the most severe disease stages.

From a clinical perspective, the consequences of misclassification during primary keratoconus screening are significant. A false-negative case, in which a mild keratoconus sample is classified as normal, is clinically more critical than a false-positive prediction because a delayed diagnosis may postpone early interventions, such as corneal collagen cross-linking, and increase the risk of progression toward advanced ectasia requiring corneal transplantation. In contrast, a false-positive case, in which a normal sample is classified as mild keratoconus, may result in unnecessary follow-up examinations and temporary patient anxiety but generally carries a lower long-term clinical risk.

The observed error distribution further demonstrated that the proposed model produced no misclassification into the advanced keratoconus category, whereas the remaining errors occurred only between the normal and mild classes. This behaviour is clinically understandable because mild keratoconus often exhibits subtle morphological variations that can partially overlap with normal corneal appearances, making early stage separation challenging, even during clinical assessment. To further reduce clinically significant false-negative predictions, future improvements may include the use of class-weighted loss functions or cost-sensitive optimisation strategies that assign higher penalties for mild keratoconus misclassification.

Table 4 presents a comparison between the proposed approach and several state-of-the-art keratoconus classification methods that rely on different imaging modalities. The KeratoDetect system introduced by Lavric and Valentin [33], which uses a VGG16-like CNN trained on synthetic Scheimpflug-derived colour-coded topography maps, reported an accuracy of 99.33% on the test set. In the study by Kuo et al. [8], multiple pretrained CNNs were evaluated on Placido-based colour-coded topography, with ResNet152 achieving the highest reported accuracy of

**FIGURE 8.** Confusion matrix illustrating the classification performance of the proposed attention-based Xception model across normal, mild, and advanced keratoconus stages.

95.80%. Wan et al. [34] developed an end-to-end pipeline combining FCN-based segmentation and a machine-learning ensemble classifier, consistently exceeding 92% accuracy when applied to corneal endothelial images captured via specular microscopy. More recently, Al-Timemyetal. [9] proposed a feature fusion approach that integrates Xception and InceptionResNetV2 representations with an SVM classifier, achieving 99.10% accuracy using multiple anterior elevation maps. In comparison, the proposed attention-based Xception model achieved an accuracy of 92.86% using only 2D RGB corneal images captured with a mobile phone, which constitutes a more challenging imaging modality but substantially reduces dependence on specialised and expensive ophthalmic equipment.

Although the proposed model does not exceed the very high accuracy figures reported for approaches based on tomographic maps or endothelial microscopy images, these methods operate on highly structured and device-calibrated inputs that inherently encode corneal geometry and morphology. These characteristics simplify the learning task and contribute to a near-ceiling performance. In contrast, the present study relied solely on raw RGB corneal photographs captured using mobile phones, which introduced greater variability in terms of illumination, resolution, and background structure. Despite these challenges, the achieved accuracy of 92.86% demonstrates that attention-enhanced CNNs can provide competitive performance while offering a practical and cost-effective alternative for preliminary keratoconus screening in environments where advanced ophthalmic imaging devices are not readily available.

A. MODEL EXPLAINABILITY USING GRAD-CAM

In this section, a Grad-CAM visualisation analysis is performed to better understand the regions of the feature maps on which the proposed attention-Xception model focuses

TABLE 4. Comparison of state-of-the-art keratoconus classification models using different imaging modalities. Only accuracy values reported in the original studies are included.

Study	Year	Model	Image Type	ACC(%)
Lavric and Valentin [33]	2019	VGG16-like CNN	Synthetic Scheimpflug colour-coded topography	99.33
Kuo et al. [8]	2020	VGG16 / ResNet152 / InceptionV3	Placido-based colour-coded corneal topography	95.80
Wan et al. [34]	2023	FCN-(ResNet50 / DenseNet121 / InceptionV3 / MobileNetV2) + PCA + ML ensemble	Corneal endothelial images (specular microscopy)	> 92
Al-Timemy et al. [9]	2023	Xception + InceptionResNetV2 + SVM	Multiple corneal anterior elevation maps	99.10
This study	2025	Attention-Xception	2D corneal RGB colour images (mobile phone)	92.86

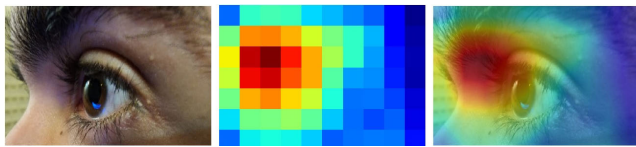


FIGURE 9. Grad-CAM visualisation of a normal corneal image (left), corresponding heatmap (middle), and overlay result (right).

during the classification process. Three test sample images were randomly selected, one from each class, and the feature maps extracted from the last convolutional layer of the proposed architecture were used to compute the Grad-CAM heat maps. The computed Grad-CAM heatmaps were resized from 10×10 to 299×299 pixels and mapped to the RGB colour space for visual superimposition onto the original corneal images.

Figures 9 to 11 present the Grad-CAM visualisations for the three classes. The Grad-CAM heatmaps use a colour spectrum ranging from dark blue to red, where dark blue and blue indicate regions with minimal contribution to the prediction, green and yellow represent moderate influence, and orange and red correspond to the regions with the strongest contribution to the final classification decision. For each class, the original test image, corresponding Grad-CAM heatmap, and overlay of the heatmap on the original image are displayed horizontally. The visualisations indicate that the proposed attention-based network consistently focused on clinically relevant corneal regions that contributed most to the classification decisions. However, the presence of imaging artefacts or longitudinal positioning variations in the test samples (e.g., Figure 9) slightly influenced the attention distribution, which may affect the classification performance in some cases.

B. COMPUTATIONAL COMPLEXITY AND INFERENCE LATENCY

The proposed attention-based modifications slightly increased the total number of trainable parameters relative to the baseline Xception architecture, from 22.1M to 24.9M parameters, corresponding to an increase of approximately

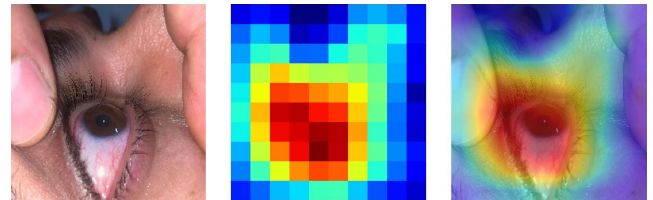


FIGURE 10. Grad-CAM visualisation of a mild keratoconus image (left), corresponding heatmap (middle), and overlay result (right).

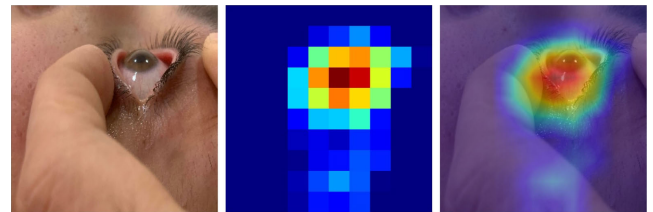


FIGURE 11. Grad-CAM visualisation of an advanced keratoconus image (left), corresponding heatmap (middle), and overlay result (right).

12.7%. However, this increase in model complexity was accompanied by an improvement of approximately 5% in the classification accuracy. Compared with larger backbone architectures, such as InceptionResNetV2 and ResNet152, the proposed attention-Xception model achieved both higher classification performance and substantially lower parameter counts, reducing the number of parameters by approximately 54.9% and 58.2%, respectively.

To further evaluate the computational efficiency of the proposed architecture, inference latency experiments were conducted on both a desktop-grade CPU platform and a low-power embedded device representative of portable clinical systems. The average inference times were measured using 10 test images with a resolution of $299 \times 299 \times 3$. The experiments were performed on an Intel Core i7-1165G7 CPU and an NVIDIA Jetson Nano embedded AI platform. The Jetson Nano used CUDA-enabled GPU acceleration during inference, which contributed to the reduced latency relative to CPU-only execution. Table 5 summarises the average inference times and the corresponding standard deviations.

TABLE 5. Average inference time comparison (mean $\pm$ standard deviation in seconds) using 10 test images.

Processor	Xception	InceptionResNetV2	ResNet152	Proposed Model
Intel Core i7-1165G7 CPU	0.45 $\pm$ 0.13	0.81 $\pm$ 0.23	0.83 $\pm$ 0.23	0.52 $\pm$ 0.14
NVIDIA Jetson Nano	0.28 $\pm$ 0.09	0.48 $\pm$ 0.14	0.49 $\pm$ 0.14	0.31 $\pm$ 0.09

TABLE 6. Overall performance metrics from the ablation study of different architectural configurations of the proposed attention-Xception model.

Model Setup	Overall Performance Metrics			
	ACC (%)	PRE (%)	REC (%)	F1 (%)
Attention + Concat + 16 Blocks	92.86	92.98	92.86	92.85
Attention + Concat + 8 Blocks	92.09	92.22	92.10	92.11
Attention + Add + 16 Blocks	90.96	91.02	91.23	90.95
Attention + Add + 8 Blocks	90.27	91.38	90.76	90.55
No Attention + Add + 16 Blocks	88.10	88.23	88.10	88.09
No Attention + Add + 8 Blocks	88.21	88.30	88.16	88.22

TABLE 7. Overall performance metrics and inference latency comparison for different numbers of attention heads.

Attention Heads	Overall Performance Metrics				Inference Latency	
	ACC (%)	PRE (%)	REC (%)	F1 (%)	CPU (s)	GPU (s)
1	92.86	92.98	92.86	92.85	0.5200 $\pm$ 0.0470	0.0236 $\pm$ 0.0020
2	92.90	93.02	92.89	92.94	0.5832 $\pm$ 0.0543	0.0265 $\pm$ 0.0024
4	92.79	92.67	92.83	92.62	0.6437 $\pm$ 0.0674	0.0292 $\pm$ 0.0029

TABLE 8. Overall performance metrics from the 42-fold cross-validation evaluation of the proposed attention-Xception model (mean $\pm$ standard deviation).

Overall Performance Metrics			
ACC (%)	PRE (%)	REC (%)	F1 (%)
92.77 $\pm$ 0.47	93.01 $\pm$ 0.47	92.85 $\pm$ 0.48	92.85 $\pm$ 0.47

The results demonstrate that the proposed model maintains a relatively low inference latency despite the additional attention operations. On the Intel Core i7-1165G7 CPU, the proposed architecture achieved an average inference time of 0.52 $\pm$ 0.14 s per image, whereas inference on the NVIDIA Jetson Nano required 0.31 $\pm$ 0.09 s per image. Although the proposed model is slightly slower than the baseline Xception network because of the added attention computations, it is substantially faster than larger architectures such as InceptionResNetV2 and ResNet152.

These inference times are suitable for computer-aided diagnostic support systems, where keratoconus staging does not require strict real-time processing constraints. Furthermore, additional optimisation strategies, such as FP16 or INT8 quantisation and structured pruning, may further reduce memory usage and inference latency for deployment on portable or resource-constrained medical devices.

C. ABLATION STUDY

An ablation study was conducted under six different architectural settings (Table 6) to evaluate the individual contribution of each component of the proposed attention-Xception model. Initially, the effect of extending the middle-flow depth from 8 to 16 blocks was evaluated. The results show that increasing the middle-flow depth improves the classification performance when attention mechanisms are used, resulting in an improvement of approximately 0.8% across most evaluation metrics. In contrast, when attention modules are not employed, extending the middle flow provides little or no noticeable improvement in the performance.

The contribution of the attention mechanism was investigated by comparing models with and without attention layers. The results indicate that the inclusion of attention improves the overall classification performance by more than 4% across all metrics. Finally, the proposed feature fusion strategy was evaluated by replacing the channel-wise concatenation with the conventional element-wise addition. The results demonstrate that when additive fusion is used with the attention mechanism, the classification performance decreases by approximately 1%, suggesting that feature concatenation is more effective at preserving complementary feature information.

In addition to the architectural ablation analysis, further experiments were conducted to investigate the effect of

different numbers of attention heads on both classification performance and inference latency. Table 7 presents the classification performance and inference latency of the proposed model using different numbers of attention heads. The results indicate that changing the number of attention heads produced only marginal differences in the classification metrics. Although the two-head configuration achieved the highest numerical values, the performance differences among the tested settings were not substantial. The limited performance variation observed across the tested attention-head configurations suggests that the discriminative information available in the RGB corneal images can already be effectively captured using a small number of attention heads. Consequently, increasing the number of heads increases the computational overhead of the attention mechanism without providing a proportional improvement in classification performance. Therefore, the single-head configuration was retained in the proposed architecture because it offers a favourable trade-off between classification performance and computational efficiency.

The inference latency results further support this observation. It was found that increasing the number of attention heads slightly increased the inference latency on both the CPU and GPU platforms because of the additional attention computations and memory access requirements. However, the increase remained relatively small across all tested settings. The reported inference times represent the average latency per image computed using ten test images with a resolution of $299 \times 299 \times 3$. GPU-based inference was performed on an NVIDIA Jetson Nano using CUDA-enabled acceleration.

D. ADDITIONAL CROSS-VALIDATION EVALUATION

In addition to the learning curves (training/validation loss and accuracy over epochs), which demonstrate the stability and generalisation capabilities of the proposed model, an additional k -fold cross-validation analysis was conducted to further evaluate the robustness and variability of the architecture across different testing folds. The dataset consisted of 420 images and was divided into 42 folds ($k = 42$), with each fold containing 10 images. For each iteration, one fold was used for testing, while the remaining 41 folds were used for training, resulting in 42 independent evaluation runs in which a different fold was selected for testing each time.

The data augmentation procedure described in Section VI-A was applied only to the training folds after the train-test split was performed during each iteration to ensure that augmented variants of the testing images were not introduced into the training dataset. The mean and standard deviation of the evaluation metrics obtained across all 42 folds are summarised in Table 8. The averaged results further indicate the consistency and generalisation capability of the proposed attention-Xception model beyond the fixed train-test split used in the primary experiments. In addition, the relatively low standard deviation values indicate a stable performance and low variance across different testing folds.

VIII. CONCLUSION

This paper presents an attention-enhanced CNN based on a modified Xception architecture for automated keratoconus staging using 2D RGB corneal images. The proposed framework integrates depthwise-attention blocks, extended middle-flow depth, and channel-wise feature fusion with normalisation to improve cross-channel interactions and preserve subtle corneal patterns during hierarchical feature extraction. These architectural modifications were specifically designed to address the challenges of medical image analysis, in which fine-grained morphological differences distinguish disease stages. The experimental evaluation demonstrated that the proposed attention-Xception model achieved an overall classification accuracy of 92.86%, with balanced precision, recall, and F1-scores exceeding 92% across the normal, mild, and advanced classes. Compared with the standard Xception backbone, which achieved an accuracy of 88.10%, the proposed architecture provides a relative improvement of 4.76% while increasing the parameter count by only 2.8 million (from 22.1M to 24.9M). This indicates that the observed performance improvement is based on architectural refinements rather than increasing the model capacity. Unlike most studies which rely on corneal tomography or topography acquired using specialised and expensive imaging devices, this study uses low-cost equipment (*i.e.*, a smartphone's camera). This feature significantly expands the applicability of automated keratoconus screening, particularly in resource-constrained environments where access to advanced ophthalmic devices is limited. The results demonstrate that attention-guided CNNs can deliver reliable stage-wise discrimination from non-standardised images and offer practical, accessible, and scalable solutions for the early detection of keratoconus.

Although the proposed method achieved promising results, the dataset size remained relatively limited, which may affect the generalisation capability of the model in broader clinical settings. In addition, variations in image quality, illumination conditions, patient positioning, and acquisition devices may influence classification performance. Furthermore, the dataset was collected under a limited clinical setting, and additional multicentre datasets containing more diverse demographic and imaging characteristics are required to further validate the robustness and generalisability of the proposed architecture. Therefore, future work will focus on collecting larger and more diverse datasets, improving cross-domain robustness, and enhancing the model using advanced deep learning and data augmentation techniques.

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